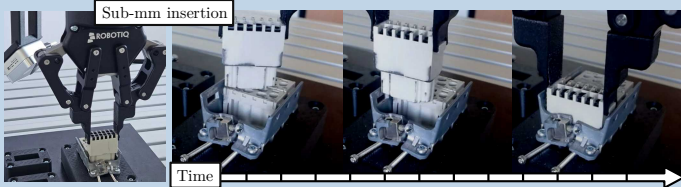


Sub-mm insertion



Time

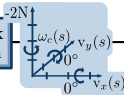
Environment

Low-dim Task
Frame Action

Impedance

$\text{clip}(\mathbf{F}, -\mathbf{F}_{lim}(o_t), \mathbf{F}_{lim}(o_t))$

$\mathbf{F}_t$



$\mathbf{a}_t$

10 Hz

$\mathbf{o}_t, r_t$

Learner

Human
Buffer

Policy
Buffer

Actor
 $\pi_\theta(s)$

Critic
 $Q_\phi(s, a)$

Transitions

Parameters

Actor

Human
Intervention

RL Policy
 $\pi_\theta(s)$

$\mathbf{a}_{ITV}$ or $\mathbf{a}_{RL}$

Structure

AMP2:Insert

$\lambda(\mathbf{o}_t)$

MP3:Press

$p_z \leq 0.5\text{mm}$

$\mathbf{F}_z \geq 2\text{N}$

MP1:Approach